

Comparative Delay, Noise and Energy of High-Performance Domino Adders with Stack Node Preconditioning (SNP)

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Abstract

Stack node preconditioning (SNP) and “mutex” techniques for charge-sharing noise reduction are incorporated into the critical path gates containing transistor stacks in 32-bit Domino adders to simultaneously improve best achievable performance by 10% and reduce charge-sharing noise by 2X in circuits containing transistor stacks.

I. Introduction

Domino logic circuits have worse noise margin than static CMOS logic since both dynamic and static stages in a domino gate are skewed strongly in the direction of “evaluate” transitions to maximize performance. Domino still remains the circuit style of choice for high-performance ALU designs since it provides a favorable trade-off between performance and noise margin where significant performance gain is achieved at reduced, but adequate, noise immunity. This trade-off, however, becomes less favorable with technology scaling since the noise margin of domino circuits degrades more severely than that of static logic as transistor threshold voltage (V_T) is reduced in the same proportion as supply voltage (V_{cc}) [1]. Noise margin considerations thus limit the scalability of V_T in domino adder designs, and degrade V_{cc}/V_T ratio [2]. The resulting deterioration of transistor performance is even worse in circuits containing transistor stacks because of body-effect induced V_T increase. Circuits with transistor stacks are also susceptible to additional noise induced by charge sharing between the stack node and the output node. Consequently, both performance and noise immunity degrade significantly in a 32-bit Kogge-Stone adder that serves as the basis of ALU architectures in high-performance processors and whose critical paths consist mainly of circuits with transistor stacks.

In this paper, we present a technique – referred to as SNP – to improve the worst-case “evaluate” delay of domino circuits with transistor stacks by pre-conditioning the stack node to a desired voltage during the precharge phase. SNP is combined with the “mutex” or “cross-connected inputs” configuration of transistor stacks [3] to simultaneously improve performance and reduce charge sharing noise. Direct output predischARGE (DOP) is incorporated in standard domino gates to allow further delay improvement, subject to noise margin constraints. Delay, noise and energy trade-offs offered by SNP, “mutex” and DOP are compared with standard domino and with an alternate stack node preconditioning technique which eliminates charge sharing noise. These techniques are incorporated in different implementations of a 32-bit Kogge-Stone adder to determine the most energy-efficient, high-performance and robust design.

II. Delay, Noise and Energy Comparisons

DOP (Fig. 1b) and SNP (Fig. 1c) are incorporated in the standard D2 domino (Fig. 1a) implementation (no clocked foot NMOS) of a basic “generate” cell in tree-type adders such as Kogge-Stone. Since the critical path of the adder is essentially a cascade of this basic cell, it is used to illustrate the trade-offs in delay, noise and energy for DOP, SNP and “mutex” at the logic gate level. Simulations show that DOP improves the best achievable “evaluate” delay by 5% since the static stage can be skewed more in favor of the “evaluate” transition without degrading precharge time. However, additional short-circuit current through the static stage during precharge, and additional capacitances from the predischARGE NMOS as well as its clock driver cause the switching energy consumption of DOP to be larger. Domino with DOP will be used as the reference case for delay, noise and energy comparisons in the rest of the paper.

Delays of the dynamic and static stages in the basic cell depend strongly on the relative arrival times of the inputs to the stack transistors. Let us consider the dynamic and static gate delay of the DOP implementation as shown in Fig. 2. When input A# (A) of the stack transistor close to the output node arrives much after the input B# (B) of that away from the output, the delay T_F is 20% to 30% smaller than the delay T_S when A# (A) arrives much earlier than B# (B). The delay improves because the stack node is automatically preconditioned by input B# (B) to (1) virtually eliminate V_T increase from the body effect, and (2) reduce voltage swing of the stack node during output transition. The gate delay is worst when inputs to the stack transistors arrive simultaneously. This delay T_W is larger than T_S because of reduced gate drive for the stack transistor close to the output during the initial phase of the output transition. T_W is also larger than T_F because, in the worst case, the stack node is preconditioned by inputs during the previous cycle to cause V_T increase from body effect and maximum voltage swing at the stack node. The largest delay improvement due to SNP occurs when input A# (A) arrives 15ps before input B# (B). In this case, the early arriving input causes the output voltage to fall slightly due to charge sharing before the input B# (B) arrives to complete the transition. Inputs to the stack transistors of the basic cell in the critical path of a Kogge-Stone adder arrive very close to each other because of the balanced adder topology. Therefore, performance of the adder is dictated by the worst-case delay T_W . We use SNP to always predischARGE the NMOS stack node of the dynamic stage to ground, and precharge the PMOS stack node in the static stage to V_{cc} during the precharge phase so that the worst-case “evaluate” delay of the basic cell approaches T_F in

the design window of interest. As shown in Fig. 2, SNP improves the delay T_w by 25% compared to DOP (or standard domino). The delay converges to T_F , as expected, when A# (A) arrives well after B# (B). If A# (A) arrives much before B# (B), SNP delay approaches T_S since the stack nodes get preconditioned through the PMOS keeper in the dynamic stage and the pull-down NMOS in the static stage, in a direction opposite to that achieved by using SNP. Since the time constant for this preconditioning is rather large (~100ps), the performance gain of SNP is preserved even if A# (A) arrives slightly before B# (B).

Charge sharing noise, inherent in circuits with transistor stacks, can be improved by splitting the transistor stack into two halves and by transposing the order of the stack transistor inputs from one half to the other so that, in the worst case, charge sharing occurs between the output node and only one of the split stack nodes [3]. This technique, referred to as “mutex” in this paper, is incorporated into DOP (Fig. 3a) and SNP (Fig. 3b). An alternate technique to eliminate charge sharing noise is shown in Fig. 3c, where the NMOS stack node in the dynamic stage is precharged to $V_{cc}-V_T$, and the PMOS stack node in the static stage is predischarged to V_T during the precharge phase. Noise simulation results (Fig. 4) show that charge sharing noises are identical in standard domino and DOP. The noise is slightly larger in SNP because of additional capacitance at the stack nodes introduced by the transistors used for stack node preconditioning. The noise is reduced by approximately 2X when “mutex” is incorporated into DOP or SNP, and is completely eliminated when the stack nodes are preconditioned, as in Fig. 3c, in a direction opposite to that of SNP.

Gate delays for different arrival times of inputs to the stack transistors are shown in Fig. 5 for the basic cell where charge sharing noise techniques have been incorporated into DOP and SNP. For the “DOP+mutex” implementation, only one of the split stack nodes gets preconditioned in the direction which causes V_T and stack node swing to increase, regardless of whether A# (A) arrives well before B# (B) or well after or at the same time. As a result, delay is expected to improve from that in DOP when A# (A) arrives much before or at the same time as B# (B), and should degrade when A# (A) arrives much later. This would indeed be the case for simple stacks containing transistors of equal widths. However, for the basic cell being considered here, output load capacitance increases when half of the two transistors in parallel at the bottom of the stack are moved to the top in the “mutex” configuration. Consequently, the expected improvement in delay T_S or T_w is nullified, and the delay T_F is further degraded. Fig. 6 shows total transistor width vs. worst-case delay (T_w) comparisons for different implementations of the basic cell. Device widths are scaled over a large range without changing the relative width ratios to trade-off delay vs. energy consumption. The total of transistor widths in a gate, which approximately tracks the total switched capacitance, is used as a measure of energy consumption throughout the paper. Delay behavior is virtually unchanged from that of DOP (Fig. 5) when stack nodes are

preconditioned for zero charge sharing noise. In view of the diminishing performance improvement in domino adders with technology scaling, preconditioning the stack nodes for zero charge sharing noise – but no performance improvement – does not provide a favorable trade-off between noise and performance, especially if charge sharing noise can be reduced sufficiently to satisfy the overall noise margin requirement of the gate. While “DOP+mutex” reduces charge sharing noise significantly, it does not offer any improvement in the worst-case delay either. The “DOP+SNP+mutex” implementation, on the other hand, simultaneously improves both charge sharing noise and the worst-case delay. Best achievable delay (Fig. 6) is 10% better than DOP, and 14% better than “DOP+mutex”. Energy consumption at equal delay is also smaller. At the same time, charge sharing noise is comparable to that in “DOP+mutex”. “DOP+SNP” improves the best achievable delay by another 16%.

III. Kogge-Stone Domino Adder with DOP, SNP and Mutex

Critical path delay and energy dissipation of two different implementations of a 32-bit Kogge-Stone domino adder (Fig. 7 & 8) are compared. DOP is incorporated in all gates. SNP is applied to all D2 dynamic stages and static stages containing NMOS or PMOS stacks. The critical path of the first implementation consists of 13 stages (Fig. 7a) in which every static stage is an inverter (Fig. 7b). The critical path of the second implementation contains 7 stages (Fig. 8a) where “propagate (P) + generate (G)” (PG) logic is performed in both dynamic and static stages (Fig. 8b). While the critical path of the 13-stage implementation has only NMOS stacks in the G sub-cell of the basic PG cell, the 7-stage version contains both NMOS and PMOS stacks. The basic PG cells in both of these versions (Fig. 7b & 8b) use complementary input/output, instead of true input/output, to eliminate transistor stacks in the P sub-cell which drives larger fanout than the G sub-cell. As a result, critical path delay improves by 5% to 10%, compared with an implementation that uses true input/output. As shown in Fig. 9, the best achievable critical path delays are virtually identical in both 7-stage and 13-stage implementations of the adder. Potential improvement in the performance of the 13-stage adder from elimination of PMOS stacks is nullified by approximate doubling of the number of stages in the critical path. In addition, the 13-stage implementation dissipates 30% more energy than the 7-stage version for equal target delay because it contains a larger number of transistors. Using SNP improves the 7-stage adder performance by 19% and that of the 13-stage adder by 11% because while every stage in the critical path of the 7-stage version contains NMOS or PMOS stacks, the 13-stage adder has only NMOS stacks in every alternate stage. As a result, the “DOP+SNP” implementation of the 7-stage adder is 9% faster than the 13-stage version, and consumes 55% smaller energy at the equal target performance.

While the “DOP+SNP” implementation of the 7-stage adder is the fastest and most energy-efficient, it is susceptible to charge sharing noise. Incorporating “mutex” into the “DOP+SNP” implementation provides favorable trade-offs in

delay, noise and energy (Fig. 10) – the best achievable performance degrades by 9%, energy consumption at equal delay increases by 26% while charge sharing noise is reduced by 2X. Performance of the “DOP+SNP+mutex” implementation is 10% better than that of the implementation with DOP or “DOP+mutex” or the stack preconditioning technique that eliminates charge sharing noise. Also, energy consumption for the same performance is reduced by 44%. At the same time, charge sharing noise is 2X smaller than in the DOP implementation, and is comparable to that for “DOP+mutex”. Simulations also show that delay, noise and energy advantages offered by “DOP+SNP+mutex” are maintained as V_{cc} is scaled down.

IV. Conclusions

A stack node preconditioning (SNP) technique and “mutex” configuration of stack transistors are incorporated into DOP

implementations of domino circuits with transistor stacks to simultaneously improve the worst-case delay of the basic cell in the critical paths of 32-bit Kogge-Stone adders by 10% without any increase in energy consumption, and to reduce charge sharing noise by 2X. Best achievable critical path delay of the adder improves by 10%, and energy consumption at equal performance is 44% smaller. If charge sharing noise is not a major concern, then adder performance can be improved by 19%, instead of 10%, by using only SNP. Thus, these delay and noise improvement techniques are shown to offer favorable trade-offs in delay, noise and energy.

References

- [1] V. De, S. Borkar., *ISLPED'99*, pp. 163-168
- [2] S. Thompson *et. al.*, *1997 Symp. VLSI Tech.*, pp. 69-70
- [3] D. Allen *et al.*, *1999 ISSCC*, pp.438-439

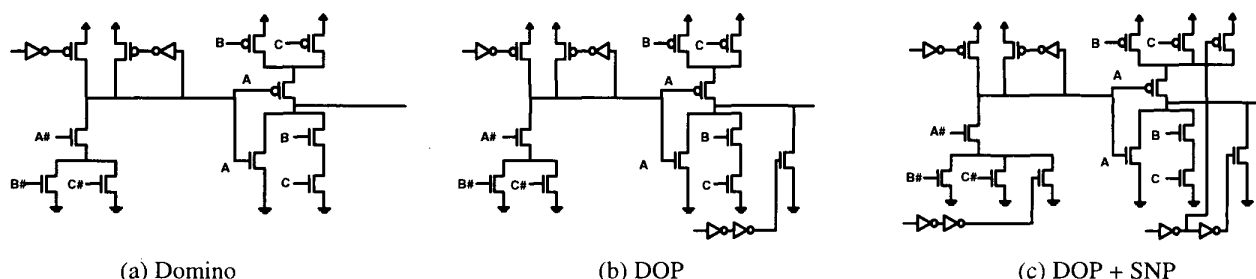


Fig.1: Circuit schematic of a domino gate with direct output precharge (DOP) and stack node preconditioning (SNP).

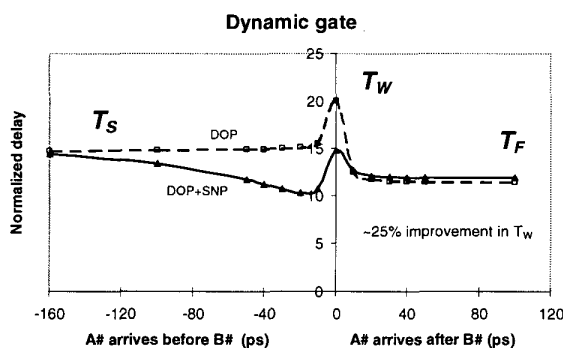


FIG. 2(a): Dynamic stage delay of DOP and SNP when inputs arrive at different times.

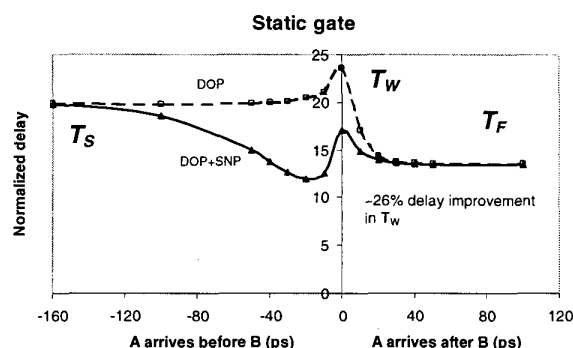


FIG. 2(b): Static stage delay of SNP and DOP when inputs arrive at different times.

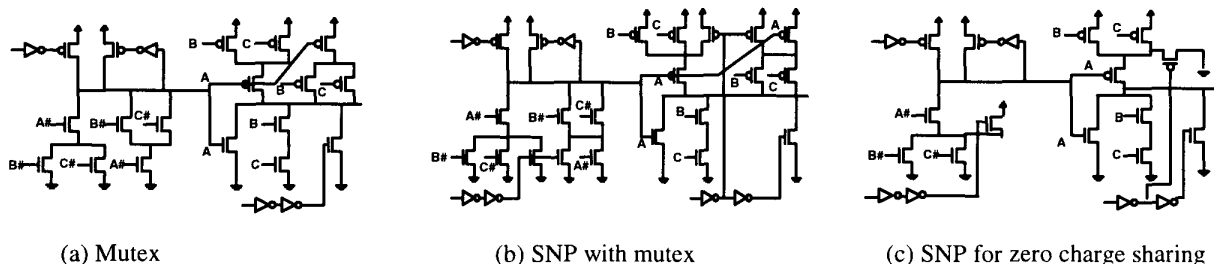


Fig. 3: Circuits with charge sharing noise reduction techniques – “mutex” and SNP for zero charge sharing.

Circuit	Dynamic gate		Static gate	
	Noise amplitude (mV)	Noise pulse width (ps)	Noise amplitude (mV)	Noise pulse width (ps)
DOP	242	102	312	70
DOP+SNP	288	111	317	73
DOP+mutex	104	116	167	76
DOP+SNP+mutex	156	119	173	80
DOP+SNP for noise	0		0	

Fig. 4: Charge-sharing noise in dynamic and static gates of the different circuit configurations

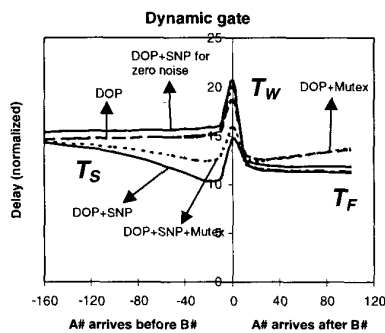


Fig. 5(a): Dynamic gate delays for different arrival times of inputs.

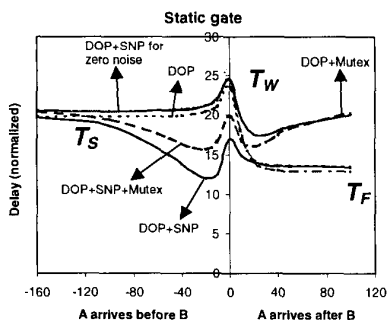


Fig. 5(b): Static gate delays for different arrival times of inputs.

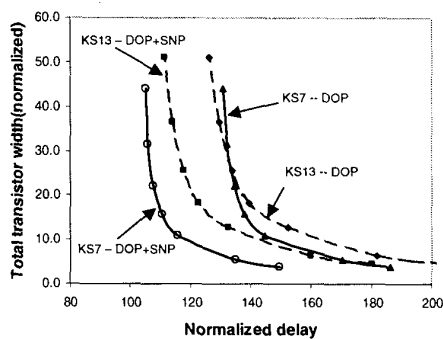


Fig. 9: Comparison of total transistor width vs. delay for the 13-stage and 7-stage Domino adder implementations.

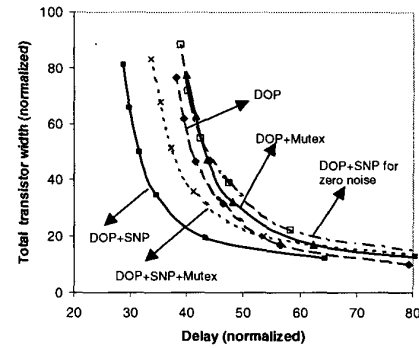


Fig. 6: Total transistor width vs. delay comparisons for different implementations of the basic cell.

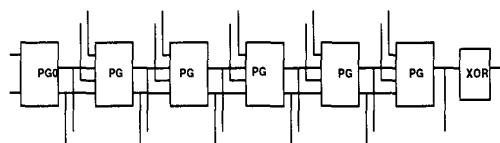


Fig. 7(a): Critical path of the 13-stage Kogge-Stone adder.

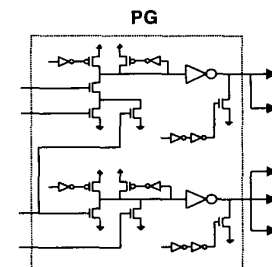


Fig. 7(b): Circuit schematic of the basic cell in the 13-stage implementation.

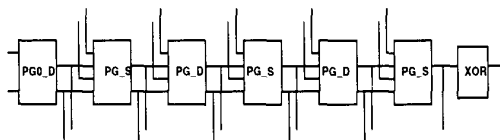


Fig. 8(a): Critical path of the 7-stage Kogge-Stone adder.

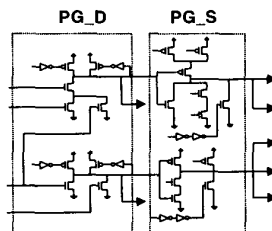


Fig. 8(b): Circuit schematic of the basic cell in the 7-stage implementation.

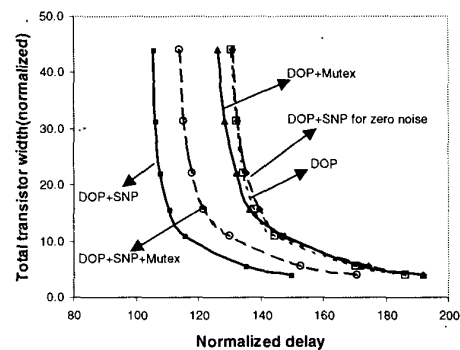


Fig. 10: Comparison of total transistor width vs. delay trade-offs of the 7-stage Kogge-Stone Domino adder using the different circuit techniques.